Message Bus Throughput Plan

Q3 2026 event forecast | Owner: Drew Rowan, Platform SRE | Next review: 2026-08-14

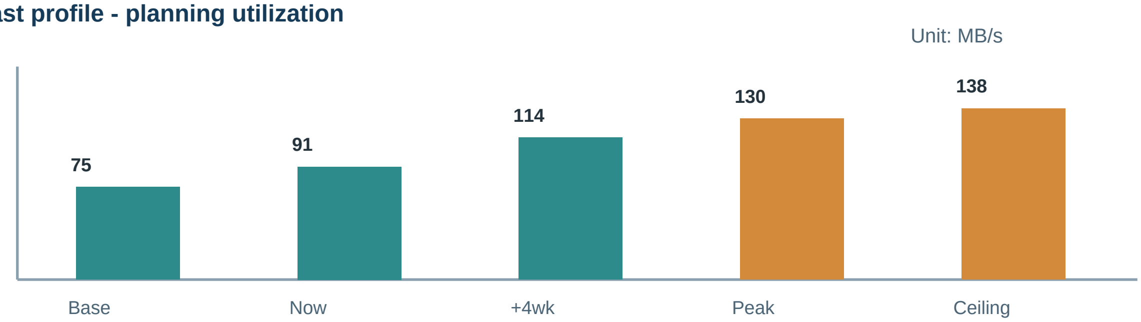
Decision at a glance

Decision	Planning peak	Evidence
Increase replication bandwidth and pre-create three settlement partitions. Throughput capacity is credible only if skew stays below 1.50x; broker count alone would not correct a concentrated settlement key.	138 MB/s	Moderate; settlement replay and broker telemetry aligned

Scope and assumptions

The bus plan models fare-settlement events, payment status fan-out, and delayed consumer catch-up. It assumes three new partitions, replication factor unchanged, and a compressed payload median of 1.8 KB. Notification topics and audit replay are intentionally excluded because their retention is owned elsewhere.

Forecast profile - planning utilization



Capacity signals

Measure	Current	Planning limit	Decision signal
Broker ingress	91 MB/s	138 MB/s	Pre-create partitions at 118 MB/s sustained.
Consumer lag	42 s	< 120 s	Lag is the customer-facing guardrail during settlement.
Replication egress	176 MB/s	< 245 MB/s	Raise broker bandwidth before replica saturation.
Partition skew	1.32x	< 1.50x	Skew above target needs key distribution work.

Plan, controls, and ownership

Action plan

Action	Due	Owner	Acceptance evidence
Partition create	Jul 30	Streaming platform	Create three settlement partitions before release.
Bandwidth uplift	Aug 04	Infrastructure	Raise broker replication allowance.
Lag rehearsal	Aug 09	Payment operations	Replay peak settlement batch against synthetic consumers.
Key audit	Aug 14	Drew Rowan	Review partition-key distribution and hot shards.

Risk register

Risk	Likelihood	Impact	Response
Hot settlement key	Medium	High	Salt the synthetic settlement key before scaling brokers.
Consumer pause	Medium	Medium	Throttle producers when lag approaches 120 seconds.
Replica repair	Low	High	Reserve egress for repair during peak windows.

Decision rationale

Increase replication bandwidth and pre-create three settlement partitions. Throughput capacity is credible only if skew stays below 1.50x; broker count alone would not correct a concentrated settlement key.

Selective cross-references

Review alongside: Checkout demand forecast and Postgres storage growth plan. These linked plans are relevant because they share a limiting dependency or coordinated operating window.

Model notes: all values are synthetic planning figures. No customer records, production identifiers, secrets, or routable network addresses are included. Northstar Transit Cloud | Event capacity model